

INFO6002 DATABASE MANAGEMENT 2
ASSIGNMENT 2

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According to Business Dimensional Lifecycle Methodology by Kimball

Bus Matrix

The proposed Bus Matrix below identifies the major business processes and corresponding conformed dimensions to carry out the design of DataMart at GlobalCarHire:

Business Process	Date	Customer	Vehicle	Branch	Employee	Service	Invoice	Loyalty Tier	Extra Service
Booking	✓	✓	✓	✓	✓	✓		✓	✓
Return Finalisation	✓	✓	✓	✓	✓	✓	✓		✓
Invoicing	✓	✓	✓	✓		✓	✓	✓	✓
Service Usage Tracking	✓		✓	✓	✓	✓			✓
Customer Management	✓	✓						✓	
Fleet Management	✓		✓	✓					

Star Schema

Chosen Business Process: Booking

When a customer makes a reservation of a vehicle, the Booking process retrieves all necessary information: payment mode, service packages, additional services, and connections with the loyalty program. This operation is the background of this GlobalCarHire and perfectly suited to the fact table of star scheme.

Fact Table FactBooking

This table makes a note of all quantifiable events (facts) accessing a booking. Every row contains a single booking and holds quantitative and foreign key dimension references.

Attribute	Description
BookingID (PK)	Unique identifier for the booking
BookingDateKey (FK)	Reference to booking date in DimDate
CustomerID (FK)	Customer who made the booking
VehicleID (FK)	Booked vehicle ID
BranchID (FK)	Branch where booking was made
EmployeeID (FK)	Staff member who processed the booking
ServiceID (FK)	Service or package selected
ExtraServiceID (FK)	Optional add-ons selected (e.g., GPS, child seat)
LoyaltyTierID (FK)	Customer's loyalty level at booking time
BookingDeposit	Amount of deposit paid during booking
BookingStatus	Status of booking (Confirmed, Cancelled, Completed)
PaymentMethod	Mode of payment (e.g., Credit Card, PayPal)

Dimension Tables

All dimensions tables have descriptive attributes of the context of booking. These assist with drill-down and slicing/dicing in the BI analysis.

DimDate

- DateKey (PK)
- FullDate
- Day, Month, Quarter, Year
- DayOfWeek, WeekOfYear

Allows the trend analysis (daily, monthly, quarterly)

DimCustomer

- CustomerID (PK)
- FullName, Email, Phone
- LicenseNumber, LicenseExpiry
- MembershipStartDate
- Country, City

Allows segmenting customers and tracking the frequency of booking

DimVehicle

- VehicleID (PK)
- Make, Model, Year
- VehicleCategoryID (FK)
- SeatingCapacity, TransmissionType, FuelType
- RegistrationNumber, Status

Its application is helpful in evaluating vehicle use, trend of categories

DimBranch

- BranchID (PK)
- BranchName
- Country, City, Address
- PhoneNumber, Currency
- OperationalStatus

Enables local revenue and performance analysis and branch performance

DimEmployee

- EmployeeID (PK)
- FullName
- Role (e.g., Manager, Associate)
- EmploymentStartDate

Aids monitoring performance by employees that have made booking

DimService

- ServiceID (PK)
- ServiceType (Package/Regular)
- Description
- BasePrice, Currency
- Validity (StartDate, EndDate)

Tracks on what services/packages are selling the best

DimExtraService

- ExtraServiceID (PK)
- Name (e.g., GPS, Child Seat)
- Rate
- Description

Allows revenue tracking and additional upsells performance

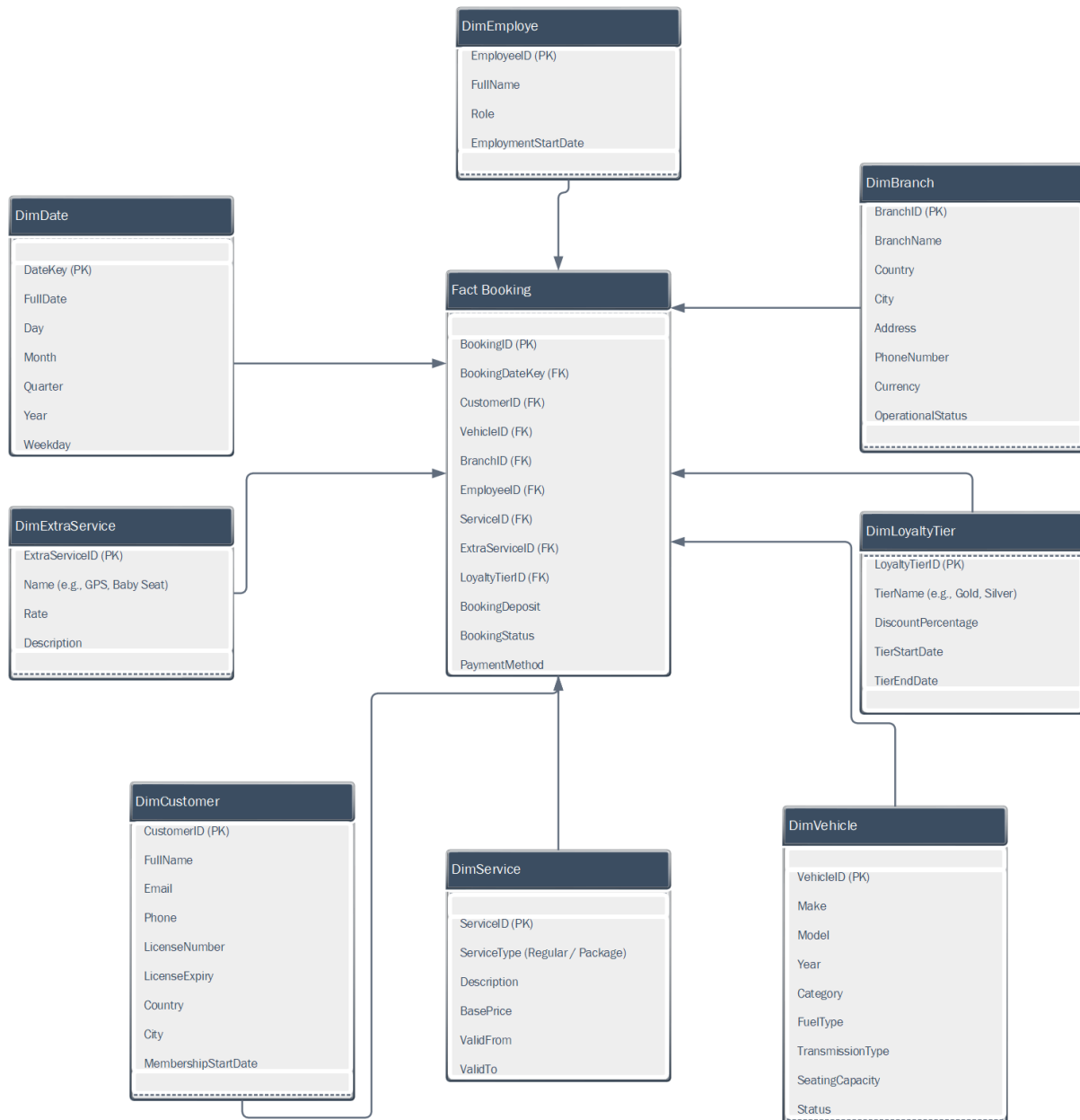
DimLoyaltyTier

- LoyaltyTierID (PK)
- TierName (Gold, Silver, Bronze)
- DiscountPercentage
- TierStartDate, TierEndDate

Allows segmenting by value and loyalty of customers

Star Schema Diagram

Booking Process: GlobalCarHire



The analysis queries

1. Branch-based Monthly Revenue Report

“Get the amount of total booking revenue that is generated by each branch sorted by month and year.”

This assists the company to know the best performing places, seasonal performance and to substitute the resources.

2. Customer Loyalty and Customer Retention Insight

“Organize customers in terms of loyalty level and indicate the number of bookings and total expenditure over the last 12 months per each customer.”

This assists in knowing the value of those that take part in such loyalty programs and aids in the personal marketing drives in order to maintain them.

3. Vehicle Utilization and Vehicle Demand Forecasting

“Find the mean number of days that every type of vehicle (e.g., SUV, Sedan) has been both booked and available, in the last quarter.”

This helps in improved management of the fleet as it shows over booked models or those that are not used much which helps in management decisions of either procuring or redistributing the vehicles.

The Datamart and Decision Making

Booking Datamart will help to make data-driven decisions on various levels of GlobalCarHire functioning. With a structure of the star schema, it concentrates on the most important business measures and dimensions, like bookings, customers, vehicles, companies, and services.

Executives and analysts can create slices and dice on any or multiple dimensions of the data - by date, location, vehicle type, level of loyalty program, or employee. This gives the organization the strength to:

- Keep track of the branch performance and assign budgets to high-growth regions
- Improve customer retention by means of segmenting loyalty and repeat booking patterns
- Maximize fleet performance, ensure efficient use of vehicles and cancel the idle costs
- Monitor uptake of the service packages, and discover upselling opportunities and customer satisfaction boosting measures
- Provide support to strategic planning by means of historical trend analysis and forecasting models

The DataMart can serve as a stepping stone to reach competitive advantage and operational excellence in a data-oriented environment by turning data into meaningful analytics as part of the operational data.

REFERENCE

Kimball, R., & Ross, M. (2013). *The Data Warehouse Toolkit: The Definitive Guide to Dimensional Modeling* (3rd ed.). Wiley.